ALGEBRA 2 · FUNCTIONS & TRANSFORMATIONS

Transformation Lab

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Transformation Lab** on Mr. K's Classroom: Algebra 2 → Functions & Transformations → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Where is the vertex of $f(x) = |x - 2| + 3$?

   (2, 3)

2. What does a negative value of a do to a graph?

   Reflects it over the x-axis.

2 Explore use the tool and record what you see

Transform the parent

Set each function and record what happened.

Function	Parent	Transformations	Key point
$g(x) = (x + 4)^2 - 1$	x^2	**left 4, down 1**	**vertex (–4, –1)**
$g(x) = -\|x\| + 2$	$\|x\|$	**reflect over x-axis, up 2**	**vertex (0, 2)**
$g(x) = \sqrt{(x - 3)}$	$\sqrt{x}$	**right 3**	**start (3, 0)**
$g(x) = \sqrt{(-x)}$	$\sqrt{x}$	**reflect over y-axis**	**start (0, 0)**
$g(x) = 2(x - 1)^3$	x^3	**vertical stretch by 2, right 1**	**center (1, 0)**
$g(x) = \frac{1}{(x - 2)} - 1$	$\frac{1}{x}$	**right 2, down 1**	**asymptotes x = 2, y = –1**
$g(x) = 2^{x+1}$	2^x	**left 1**	**(–1, 1)**

3 Write the rule fill in each blank from what you observed

1. $f(x) + k$ shifts the graph **up k units**. $f(x - h)$ shifts it **right h units**.

2. $-f(x)$ reflects the graph over the **x-axis**. $f(-x)$ reflects it over the **y-axis**.

3. $a \cdot f(x)$ with $|a| > 1$ is a vertical **stretch**. $f(bx)$ with $|b| > 1$ is a horizontal **compression**.

4. Changes inside the function act on the **x** values. Changes outside act on the **y** values.

4 Check no tool: show your work

1. Describe $g(x) = -3\sqrt{(x + 1)} - 2$.

   Reflect over the x-axis, stretch by 3, left 1, down 2

2. Write |x| shifted right 5 and down 4.

   $g(x) = |x - 5| - 4$

3. Where does the center of x^3 move in $g(x) = (x - 2)^3 + 5$?

   (2, 5)

4. Move $\frac{1}{x}$ so its asymptotes are $x = -3$ and $y = 4$.

   $g(x) = \frac{1}{(x + 3)} + 4$

5 Reflect

Why does f(x – 2) move the graph right instead of left?

Each output now happens 2 units later: the old input 0 now needs x = 2.